



Figure 10: Generated images from SMYRF-BigGAN on Celeba-HQ-256. Attention at 256×256 . The trained model uses 50% less memory compared to the memory dense attention would use.

16.1 Learning from scratch under extreme sparsity

Our initial goal was to train a SMYRF model from scratch with extreme memory reductions, e.g. to the magnitude of 99%. Such reduction could enable the training of SMYRF-BigGAN with attention at 1024×1024 . However, our preliminary experiments with BigGAN [1], failed (mode-collapse very early in the training process). We tried to investigate this further and we found that during the early stages of the training the Frobenius norm of the difference between the SMYRF and the dense attention map is really high. We believe that this is due to the non-sparsity of the attention maps in the early stages of the training. It is also possible that their eigenvalues decay slower which means that their effective rank is higher compared to pre-trained models. One way to solve the problem is to dynamically adapt the memory reduction (e.g. by selecting the number of hashes) during the training. One way to achieve that is to use as many hashes as need to achieve a certain bound for the Frobenius norm. In the early stages of training, we expect that more hashes are needed for an accurate reconstruction. The number of hashes should decay as the training progresses and the attention maps become more sparse and have lower rank. One disadvantage of this approach is that at the early stages of the training, more memory is needed. However, we observed that the period of time in which the attention maps are not very sparse is minor compared to the whole training time for BigGAN and thus this approach can lead to significant savings. We aim to explore this more in the future.

16.2 Better LSH based clustering schemes

The biggest advantage of clustering with an LSH-based scheme is that the attention complexity is linear (compared to K-means clustering for example, see Routing Transformer [19]). However, while inspecting SMYRF, we found that LSH-clustering is the biggest bottleneck to greater performances. For example, if each query attends to its' top-k (in terms of inner product) keys (instead of the keys assigned with LSH), the performance improves considerably. Finding exactly the top-k keys for each query is expensive (especially in high dimensions) and thus this approach is not viable. However, this observation motivates research in finding even more effective LSH-based clustering schemes. Even though we tried other ALSH variants, we did not manage to find something that works better than our proposed transformations till now. We consider this problem an interesting future direction since ALSH has been widely explored only for the case of a single query and multiple keys. In this paper, we did the first step in extending this to multiple queries, but we are inclined to believe that further research can lead to even better results in this direction.

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